# MILITARY SPECIFICATION

## SPECIFICATIONS, TYPES AND FORMS; MILITARY (OS) SERIES

*This specification is approved for use by all departments and agencies of the Department of Defense.*

## 1. SCOPE

1.1 This specification covers requirements for preparation of limited coordination military specifications (Military (OS) series) for the Naval Ordnance Systems Command.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein. In the event of conflict between this specification and other documents referenced herein, the requirements of this specification shall apply.

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-Q-9859 | Quality Program Requirements |
| MIL-I-45208 | Inspection System Requirements |
| MIL-S-83490 | Specifications, Types and Forms |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-414 | Sampling Procedures and Tables for Inspection by Variables for Percent Defective |

FSC MISC

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| | |
|---|---|
| MIL-STD-690 | Failure Rate Sampling Plans and Procedures |

PUBLICATIONS

Manuals

Naval Ordnance Systems Command

| | |
|---|---|
| OP 5 | Ammunitions Ashore, Handling, Stowing and Shipping |

Data Item Descriptions

Naval Ordnance Systems Command

| | |
|---|---|
| UDI-E-20602 | Test Reports, Appendant |

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply. In the event of conflict between this specification and other documents referenced herein, the requirements of this specification shall apply.

| | |
|---|---|
| DoD Manual 4120.3-M | Standardization Policies, Procedures and Instructions |
| DoD Manual 4145.26M | DOD Contractors' Safety Manual for Ammunition, Explosives and Related Dangerous Material |

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

3. REQUIREMENTS

3.1 General. Limited coordination military specifications (Military (OS) series) prepared hereunder shall be in accordance with form 1b as described in MIL-S-83490. Except as otherwise specified herein, the



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format and requirements of such specifications shall be as prescribed by the Defense Standardization Manual 4120.3-M. Figure 1 provides a sample format.

3.2 Types of specifications. The types of specifications prepared shall be in accordance with MIL-S-83490 and as specified in the contract or order.

3.3 Identification. The basic specification symbol (specification number) and the appropriate Federal Supply Classification (FSC) number will both be assigned by the procuring activity. Figure 1 illustrates the proper location for this identification information.

3.4 Coverage of specifications. Specifications may be prepared to cover a group of products, services or materials, or a single product, service or material. They are also respectively classified as either general or detail specifications.

3.4.1 General specifications. General specifications shall contain all the requirements that are common to different types, classes, grades, or styles of specific items and processes; this avoids repetition of common requirements in each detail item or process specification. It also permits changes to common requirements to be readily effected. General specifications are also used to cover requirements for designing weapons, systems and subsystems. That a document has the status of a general specification shall be indicated by its title, scope, and appropriate statements in section 3 as prescribed in Defense Standardization Manual 4120.3-M.

3.4.2 Detail specifications. A detail specification may:

a. cover all requirements for one or more types of items or services self-sufficiently so as not to be dependent on a general specification for certain common requirements, or,

b. take either the form of a specification sheet or a six-section specification, both of which are incomplete without reference to the associated general specification. The general specification (which contains the requirements common to the family of items) together with its ancillary detail specifications then constitute the total requirements. Dependent detail specifications shall not be prepared against an existing general specification unless specifically authorized by the preparing activity of the general specification.



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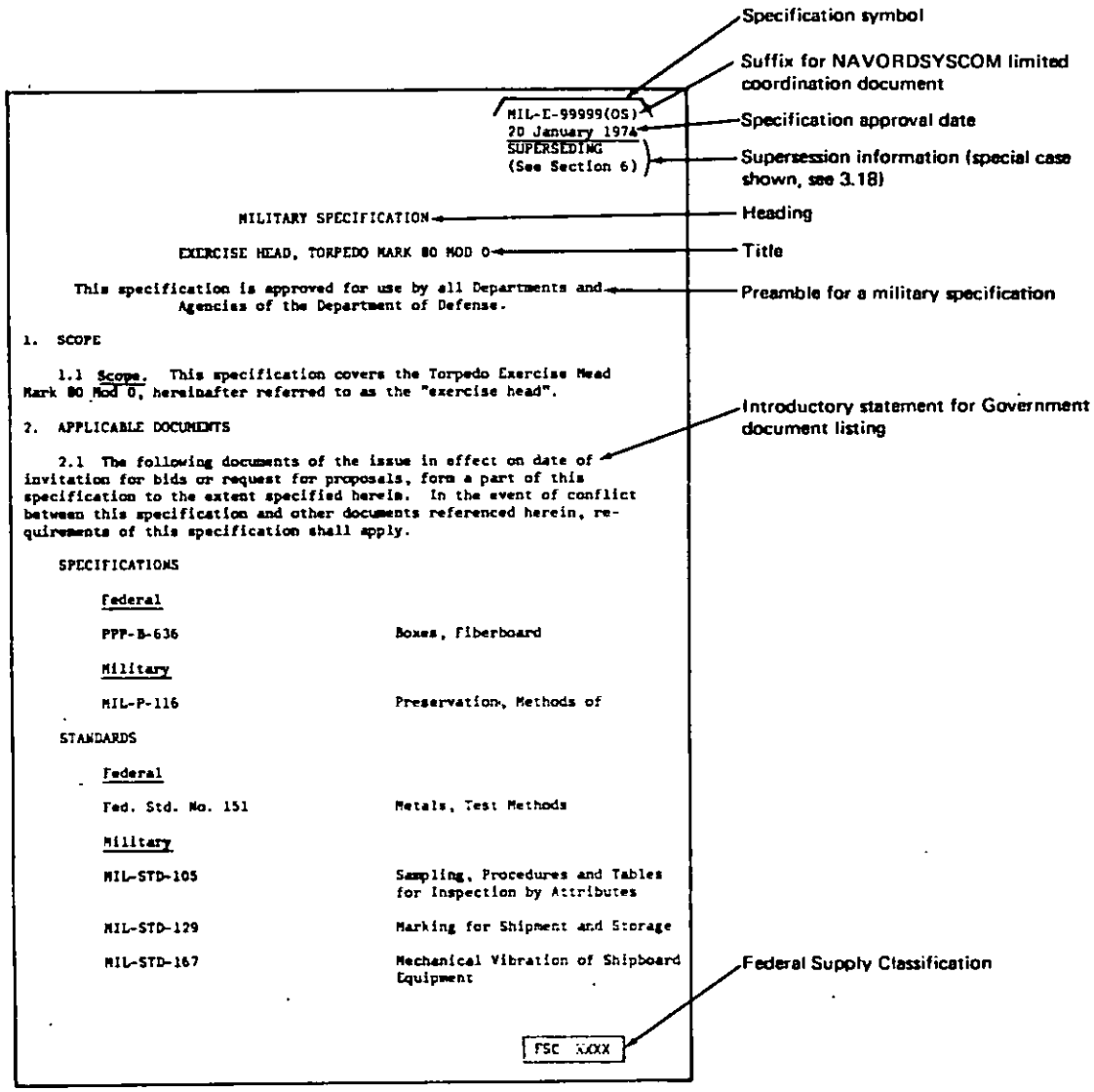



FIGURE 1. SAMPLE FORMAT FOR A MIL-(OS) SPECIFICATION



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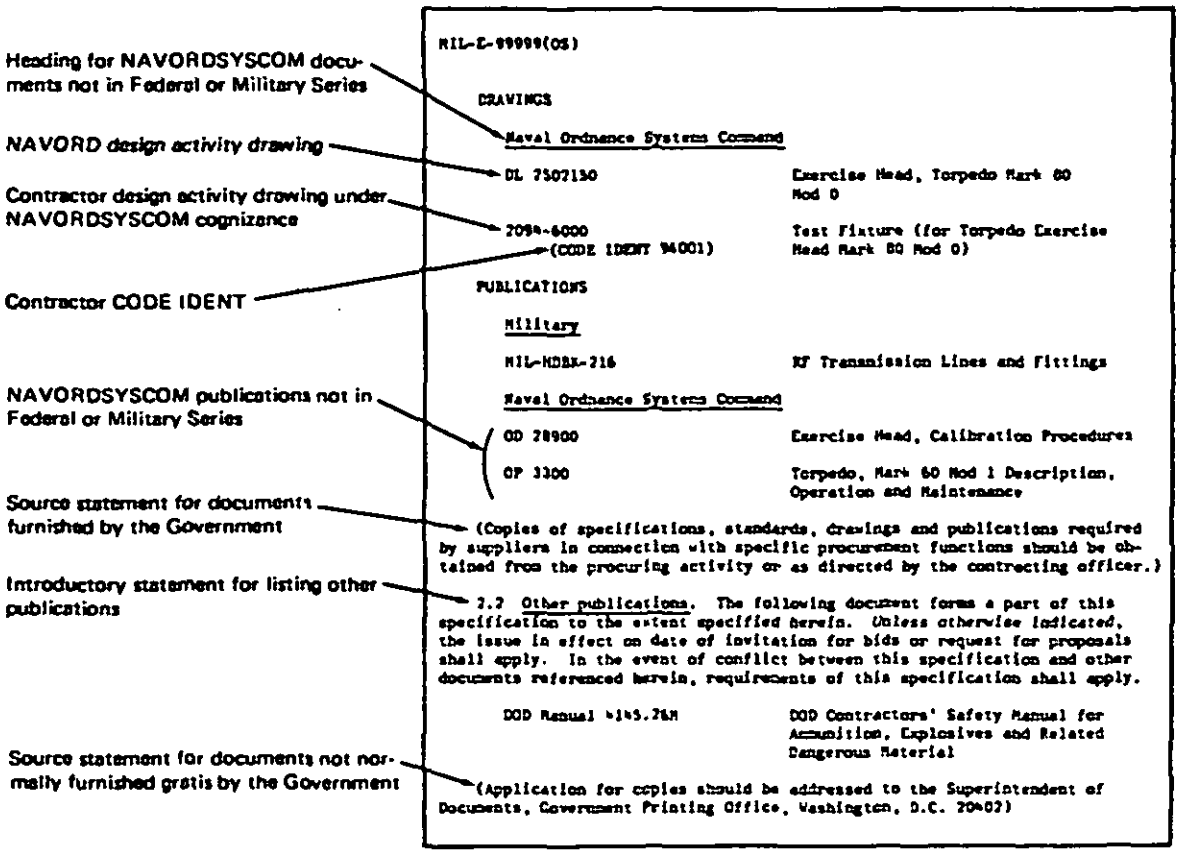



FIGURE 1.(CONTD) SAMPLE FORMAT FOR A MIL-(OS) SPECIFICATION



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3.5 Classified specifications. Each paragraph of a classified specification shall be appropriately marked using the symbols (TS) for top secret, (S) for secret, (C) for confidential, and (U) for unclassified. The symbol shall be placed immediately preceding and to the left of the paragraph involved. If all paragraphs in a classified specification have the same classification, a statement to that affect shall be included and individual paragraphs need not be marked.

3.6 Section 2 of specifications. The introductory statements in section 2 of specifications prepared hereunder shall include the requirement that, "In the event of conflict between this specification and other documents referenced herein, the requirements of this specification shall apply." All and only those documents identified and referenced in sections 3, 4, and 5 of specifications shall be listed in section 2 (see figure 1).

3.7 Use of tables. In preparing specifications, care shall be taken to provide for a systematic and concise arrangement of requirements to facilitate utilization of the specification. Requirements expressed in section 3 should be arranged in tabular form when data can thus be presented compactly and clearly. Examinations and tests of section 4 may be presented in tabular form with references to respectively applicable paragraphs of the specification. The location and preparation of tables shall be as prescribed in Defense Standardization Manual 4120.3-M.

3.8 Part numbers. New specifications for parts shall include a part numbering system in accordance with Defense Standardization Manual 4120.3-M. This system shall not be mandatory for existing or revised parts specifications.

3.9 Use of requirements stated in other documents. Requirements stated in other documents (specifications, standards, etc.) may be used and referenced in specifications prepared hereunder. However, the extent to which the requirements apply should be clearly indicated. Only documents which can readily be obtained by the specification users should be referenced. The appendix hereof lists a number of documents which have either been specifically prepared for use in weapons development or have been thoroughly reviewed and are considered suitable for use in weapons development.

3.10 Data requirements. Where applicable, provisions for procurement or delivery of data may be included in specifications as indicated in Defense Standardization Manual 4120.3-M. These contract data requirements are to be included under the heading of "6.2 Ordering data" of the



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specification, segregated into "Procurement requirements" and "Contract data requirements." Under the contract data requirements subheading, identify the requirements for deliverable data in other sections of the specification in the manner illustrated by 6.2 of this specification.

3.11 Section 4.

3.11.1 Purpose of section 4. This section shall include all of the examinations and tests (by reference where applicable) to be performed in order to ascertain that the product, material or process to be developed or offered for acceptance conforms to all the requirements listed in sections 3 and 5 of the specification. If this is not done, the obligation of the contractor and the prerogative of the Government to validate item performance may be abrogated. Conversely, section 4 shall not impose any inspection which does not correspond to a requirement listed in sections 3 or 5. In short, there shall be a one-to-one correspondence between requirements and inspections.

3.11.2 Quality conformance inspection sampling. Standard sampling procedures in accordance with MIL-STD-105, 414, 690 or other applicable standards shall be prescribed in specifications in preference to unique sampling procedures. Note that acceptable quality levels (AQL's) and other details from such standards must be stated. In certain cases 100 percent inspection may be required in lieu of sampling.

3.11.3 Qualification. Qualification requirements shall not be included in specifications prepared hereunder unless the specific circumstances described in the Defense Standardization Manual exist (see 6.4).

3.11.4 Contract quality requirements. MIL-I-45208 and MIL-Q-9858 shall not be referenced in commodity specifications. Inspection system and quality program requirements are contractual considerations and their inclusion in specifications is prohibited by Defense Standardization Manual 4120.3-M.

3.12 Appendix report form. A preprinted report form for recording results of quality conformance examinations and tests may be included as an appendix to a specification. Its use shall be limited to specifications covering complex items involving extensive testing of products. These appendixes, when used, shall conform to the sample format shown in figure 2. The report form shall be specified in sections 4 and 6 of the specification as indicated herein, citing Data Item Description UDI-E-20602 as the approved DID for acquiring the completed form. Each specification which includes such an appendix shall specify in section 4 that the appendix may be duplicated without restriction for use by testing activities.



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REPORT FORM

QUALITY ASSURANCE PROVISION

QUALITY CONFORMANCE TESTS AND EXAMINATIONS

FOR

INSTRUMENTATION ASSEMBLY

DL 2539153 (Type I), DL 2539071 (Type II),

or DL 2539544 (Type III)

(for Exercise Head Mk 85 Mod 0, Mod 1, or Mod 2)

| EQUIPMENT STATUS | | | |
|---|---|---|---|
| CONTRACTOR CERTIFICATION | | GOVERNMENT ACCEPTANCE | |
| Contractor | | Date of Acceptance | |
| Contract | | Accepted by | |
| Equipment Ser. No. | | Title | |
| Insp. Req. Satisfied | | Agency | |

FIGURE 2. SAMPLE OF APPENDIX REPORT FORM



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MIL-I-99999(OS)
APPENDIX

REPORT FORM

QUALITY ASSURANCE PROVISION

QUALITY CONFORMANCE TEST AND EXAMINATION

INSTRUMENTATION ASSEMBLY DL 2539153 (Type I),

DL 2539071 (Type II), or DL 2539544 (Type III)

10. SCOPE

10.1 This Report Form contains data sheets to record the results of the quality conformance tests and examinations of 4.7 of this specification. When completed, with authorized certification and acceptance signatures affixed, this report form or copies thereof shall accompany the instrumentation assembly bearing the recorded equipment serial number.

20. GENERAL INSTRUCTIONS

20.1 Test sequence. The quality conformance tests and examinations of 4.7 shall be completed and the results recorded or verified in this report form prior to acceptance of the instrumentation assembly. Unless otherwise noted, the sequence of testing, recording and verification, as performed in 4.7, is mandatory.

20.2 Recording of results. Where space is provided to indicate the value of specific reading obtained, the specific reading shall be recorded if a limit or limits is given in the Limit column for the test. The initials of the individual performing the inspection shall be inserted above the value observed and recorded. If limits are not stated in the Limit column, the individual performing the test or inspection shall initial in the space provided in the Reading column, indicating that the requirement has been met satisfactorily.

30. QUALITY CONFORMANCE TEST AND EXAMINATION

| | | Reference | Limits | Reading |
|---|---|---|---|---|
| 30.1 | Configuration type | 4.7 | Type I/Type II/ Type III | _____ |
| 30.2 | Test equipment | 4.7.2 | In calibration | _____ |

FIGURE 2.(CONTD) SAMPLE OF APPENDIX REPORT FORM



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| | Reference | Limits | Reading |
|---|---|---|---|
| 30.3 Test procedure | | | |
| 30.3.1 Continuity tests | 4.7.3.1 | | |
| From To | | | |
| J4806-A J4806-a | | 0.5 ohm max | _____ |
| J4806-K J4805-V, T | | 0.5 ohm max | _____ |
| 30.4 Visual inspections | | | |
| 30.4.1 Identification marking | 4.7.4.1 | | |
| 30.4.2 Interchangeability | | | |
| Diameter dimension | 4.7.4.2.3 | 11.330 in. max | _____ |
| Thickness dimension | 4.7.4.2.4 | 7.250 in. max | _____ |
| 30.5 Process verification | | | |
| 30.5.1 Lacing | 4.7.5.2 | | _____ |
| 30.5.2 Soldering | 4.7.5.3 | | _____ |
| 30.5.3 Cleanliness | 4.7.5.4 | | _____ |
| 30.6 Packaging, packing, and marking inspection | | | |
| 30.6.1 Packaging | 4.7.6.1 | | _____ |
| 30.6.2 Packing | 4.7.6.2 | | _____ |
| 30.6.3 Marking | 4.7.6.3 | | _____ |

FIGURE 2. (CONTD) SAMPLE OF APPENDIX REPORT FORM



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3.13 Safety precautions. Specifications covering explosive-loaded items, explosives, or propellants shall contain the following in the "ordering data" paragraph of section 6:

"6. The safety precaution requirements of the "Contractors' Safety Manual for Ammunition, Explosives and Related Dangerous Material" (DOD 4145.26M) are applicable.

NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunitions Ashore" (OP 5) should be made applicable."

3.14 Contract number. When the specification is prepared under contract, the contract number shall be identified in section 6 of the specification in accordance with the following sample entry:

"6.5 Contract identification. This specification was prepared by Waterhouse, Inc. and furnished to the Government under Contract No. N00156-69-C-1523."

3.15 DOD project number. The DOD project number required by the Defense Standardization Manual will be inserted by the procuring agency upon submission of the final manuscript of the specification.

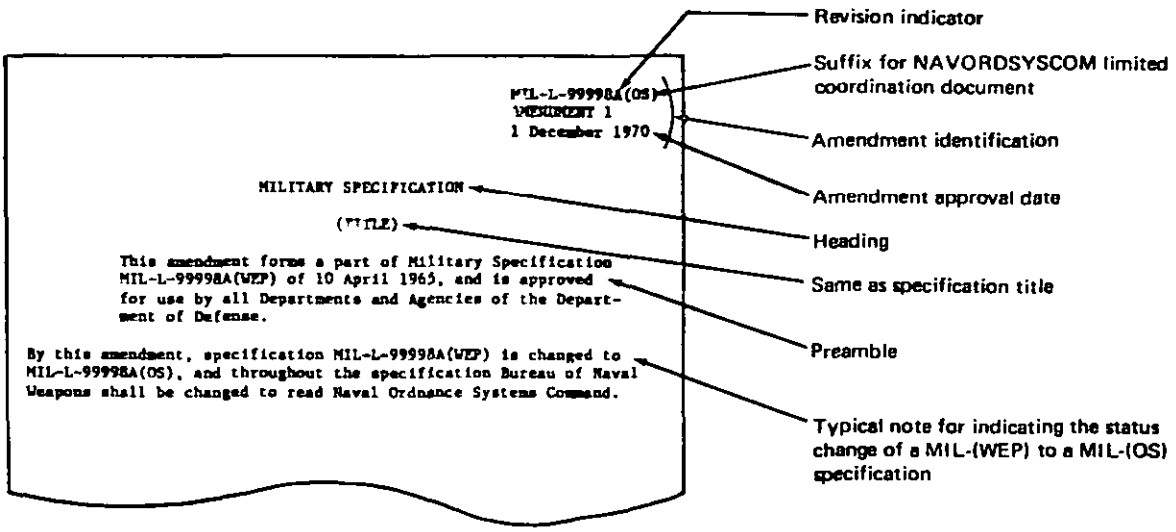
3.16 Revisions, amendments, notices, and supplements. Revisions, amendments, notices, and supplements shall be prepared in accordance with the following:

a. When revised to accomplish extensive changes, limited coordination military specifications having (WEP) and (NOrd) suffixes shall retain the same basic specification number except that the suffix will be changed to (OS). Each such revised specification shall indicate that it supersedes the MIL-(WEP) or MIL-(NOrd) specification as appropriate.

b. Amendments, notices, and supplements to limited coordination specifications in the MIL-(WEP) and MIL-(NOrd) series shall be identified with the same number and revision indicator as the basic specification, but the suffix (OS) shall be substituted for (WEP) or (NOrd). Such amendments shall conform to the sample formats in figure 3.



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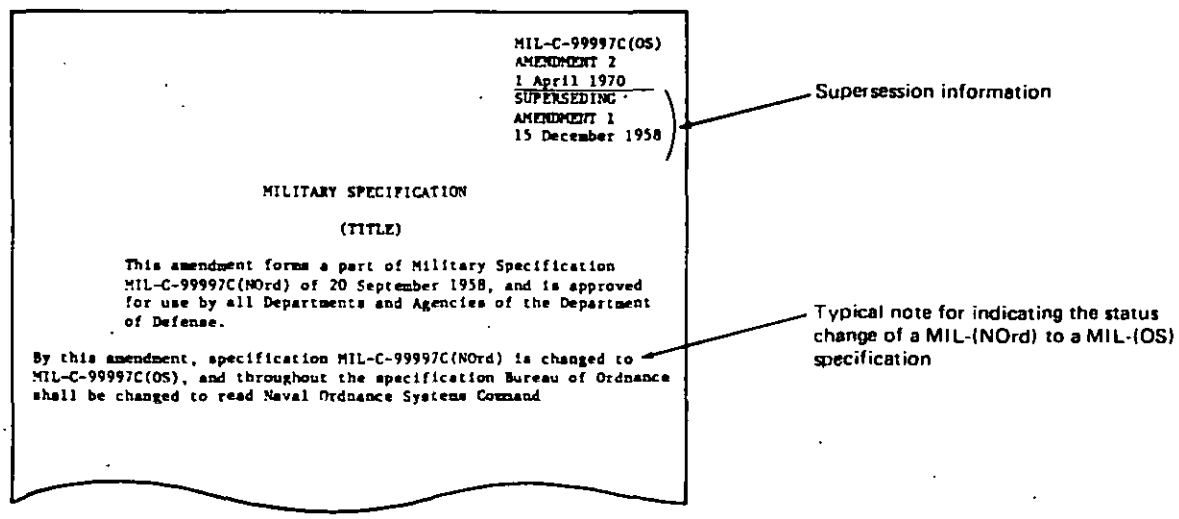




FIGURE 3. SAMPLE FORMATS FOR AMENDMENTS



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3.17 Identification of changes. Amendments and revisions of specifications shall include asterisks in the margins of the pages to indicate where changes have been made with respect to the prior issue. The following note shall be included in the NOTES (section 6) of the specification:

"6. Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue."

If the changes are extensive and too numerous to annotate, the following note will be included in the NOTES (section 6) of the specification:

"6. Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes."

3.18. Supersession of XWS, WS, or OS specifications by MIL-(OS) specifications. When a MIL-(OS) specification supersedes an XWS, WS, or OS specification, the phrase "See Section 6" shall appear on the first page under the specification symbol (see figure 1). Supersession data and any other related information shall be included in section 6; for example: "Supersession data. This specification includes the requirements of WS 9999, dated _____."

3.19 Final manuscript. Final manuscripts of specifications, revisions, amendments, supplements, and notices shall be typed single spaced on 8 by 10-1/2 inch white bond paper. The manuscript shall be of quality suitable for camera copy reproduction and be submitted undated.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance provisions shall be as specified in MIL-S-83490.



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## 5. PREPARATION FOR DELIVERY

5.1 Preparation of data for delivery shall be as specified in MIL-S-83490.

## 6. NOTES

6.1 Intended use. Limited coordination military specifications covered by this document are intended to be used for procurement of naval ordnance weapons, components, and associated supporting items for service use.

6.2 Ordering data. DD Form 1423 and other related procurment documents should specify the following:

6.2.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Type of specification (see 3.2).
c. Assignment of specification number if different from 3.3.
d. Reproducible and nonreproducible copies required (see 6.2.3).
e. Requirements for preparation for delivery (see MIL-S-83490).

6.2.2 Contract data requirements. The item of deliverable data required by this specification is cited in the following paragraph:

| Paragraph | Data requirement | Applicable DID |
|---|---|---|
| 3.1 | Specifications, Military (OS) Series | UDI-20601 |

The DID (Data Item Description/DD Form 1664) for the above data requirement is approved by the Naval Ordnance Systems Command for interim use pending inclusion in the Department of Defense Authorized Data List (TD-3). Such data will be delivered as identified on the above DID when specified on DD Form 1423 (Contract Data Requirements List) and incorporated into applicable contracts (to which the DID should be physically attached). Copies of this DID may be obtained from the Commanding Officer, Naval Ordnance Station, Standardization Division (502), Indian Head, Md. 20640.



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6.2.3 Copies of specifications. When various reproducible and non-reproducible copies of the specifications are required for delivery in addition to the final manuscript required by 3.19, MIL-D-5480 should be cited on the DD Form 1423 to the extent appropriate.

6.3 Definitions. Definitions contained in Defense Standardization Manual 4120.3-M are applicable and to be utilized as appropriate in any documents prepared within the scope of this specification.

6.4 Qualification policy and procedure. NAVORDINST 4120.2 provides uniform policy and procedures for establishing, administering, reviewing, and controlling qualified products lists (QPL's) for specifications prepared by activities of the Naval Ordnance Systems Command (see 3.11.3).

Preparing activity:
Navy - OS
Project No. MISC-N978-2



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Appendix

DOCUMENTS FOR USE IN DESIGN

10. SCOPE

10.1 Scope of appendix. The scope of this appendix is to identify documents known to be useful in the design of systems, equipments, and items for NAVORDSYSCOM.

20. Specifications for ordnance and ordnance equipment shall make use of by reference or incorporate the requirements of the following documents as appropriate:

20.1 Shipboard equipment.

| | |
|---|---|
| MIL-S-901 | Shock Tests, H.i. (High-impact), Shipboard Machinery, Equipment and System, Requirements for (Navy) |
| MIL-STD-167 | Mechanical Vibrations of Shipboard Equipment |
| MIL-STD-1399 | Interface Standard for Shipboard Systems |

20.2 Electrical and electronic equipment.

a. Design practices:

| | |
|---|---|
| MIL-STD-454 | Standard General Requirements for Electronic Equipment |

b. Selection of standard parts:

| | |
|---|---|
| MIL-STD-1631 | Procedure for Selection of Electronic and Electrical Parts during Equipment Design |

c. Physical packaging design (for ordnance equipment rather than ordnance):

| | |
|---|---|
| MIL-STD-1378 | Requirements for Employing Standard Hardware Program Modules |

16

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| | |
|---|---|
| MIL-STD-1389 | Design Requirements for Standard Hardware Program Electronic Modules |

d. Bonding and grounding:

| | |
|---|---|
| MIL-STD-1310 | Shipboard Bonding and Grounding Methods for Electromagnetic Compatibility |

20.3 Fire control equipment.

| | |
|---|---|
| MIL-F-18870 | Fire Control Equipment, Naval Ship and Shore, General Specification |

20.4 Painting of equipment.

| | |
|---|---|
| MIL-STD-1303 | Painting of Naval Ordnance Equipment |

20.5 Lubrication of equipment.

| | |
|---|---|
| NAVORD OD 3000 | Lubrication of Ordnance Equipment |

20.6 Test equipment.

| | |
|---|---|
| MIL-STD-1364 | Preferred General Purpose Electronic Test Equipment |

20.7 Selection of components/equipments.

| | |
|---|---|
| MIL-STD-143 | Standards and Specifications, Order of Precedence for the Selection of |
| MIL-STD-680 | Contractor Standardization Plans and Management |

20.8 Reliability.

| | |
|---|---|
| MIL-STD-785 | Reliability Program for Systems and Equipment Development and Production |



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20.9 Maintainability.

| | |
|---|---|
| MIL-STD-470 | Maintainability Program Requirements (for Systems and Equipments) |



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| | |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR<br>☐ USER<br>☐ MANUFACTURER<br>☐ OTHER *(Specify)*: _____ |
| b. ADDRESS *(Street, City, State, ZIP Code)* | |
| | |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |
| | |

*(TO DETACH THIS FORM, CUT ALONG THIS LINE.)*

DD FORM 02 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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